

Operator 5: Endogenous Variable Latency Calibration Operator $\mathcal{M}_{\text{latency}}$

Strict Mathematical Specification, Derivation, and Verification (Full Comprehensive Edition)

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Resource-Availability Statement This framework is built upon Status-Relational Entropy (SRE) Dynamics. All theoretical materials are archived in the Zenodo open-access repository. **This manuscript suite, including system papers, application developments, scientific hypotheses, full algebraic derivations for operators 1-6 and simulation code, is fully open-source.** Operators 7, 8, 9, 10 are subsequent closed-source commercial core modules and are not part of this manuscript suite.

Additionally, you may access the Tencent intelligent-document space supporting AI-assisted reading, which is available on both PC and WeChat mobile clients.

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- Google Gemini Notebook (historical archive, no further updates):

<https://notebooklm.google.com/notebook/ef52bf5a-f6d0-4a2a-aed4-b25d6520ab2c>

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According to the SRE principle, the physical foundation originates from information statistics.

According to the release plan laid out in the *SRE Universal Graph-Operator Pipeline & Release Roadmap*,

Operator5 is the **Endogenous Variable Latency Calibration Operator** ($\mathcal{M}_{\text{latency}}$).

This operator resides in the **Phase 2 (Causal-Blocking)** stage within the pipeline, and follows after the local-metric & probabilistic-pruning operator suite.

Its core engineering-physical mission is to perform **Relativistic Dimensional Reduction**. By computing microscopic discrete penetration rates over directed channels, it reduces Einstein's macroscopic physical phenomenon of "gravitational time dilation" down to pure algebraic measures at the graph fundamental layer. At the same time it acts as a **causal-safety interceptor** for the whole distributed streaming pipeline via logical obfuscation of Bernoulli-trial sampling routines.

I. Top-Level Algebraic Definition and Design Philosophy

Within Status-Relational-Entropy (SRE)-dynamics networks, spacetime is not a hard-coded background manifold. Instead it is an emergent endogenous metric driven by causal-stream propagation under discrete-pulse iterations (step index n). Nevertheless, as local network topological density expands inhomogeneously, the actual pulse-step cost for information streams to traverse topological sub-domains of differing density undergoes adaptive distortion. Without proper calibration, causal-time axes across distributed actors become logically misaligned and trigger cascading control collapse.

Operator 5 ($\mathcal{M}_{\text{latency}}$) defines exactly this “inhomogeneity of time-flow velocity” as **endogenous variable latency over directed channels**:

- **Domain (Input)**: Graph-edge output weights $W_e(i, j)$ computed by upstream Operator 4 (encoding local topological-overlap density), together with current microscopic endogenous relaxation step s .
- **Codomain (Output)**: Algebraic invariants defined on the graph are strictly mapped onto the **Discrete Penetration Rate** $c_e^{(s)}$ for every directed edge:

$$\mathcal{M}_{\text{latency}} : \mathbb{R}^{n \times n} \times \mathbb{N} \longrightarrow \mathbb{R}^{|E_n|}$$

II. Mathematical Derivation, Closed-Form Construction and Saturation-Clamping for Discrete-Penetration-Rate Equation

2.1 Introduction of Core Topological-Density Weight

According to the universal physical representation mapping (Theorem 6.1), when causal streams cross regions of high topological density (macroscopic massive-source regime), their superimposed core weight W_e expands exponentially. To reproduce this impedance-originated non-linear warping and conduction retardation inside a discrete algebraic space, a logarithmic adaptive contraction operator must be constructed.

2.2 Closed-Form Construction and Rigid Hardware-Level Clamping for Microscopic-Penetration-Rate Equation

Within flat or sparse vacuum regimes ($W_e \rightarrow 0$), $\ln(1 + W_e) \rightarrow 0$ would induce division-by-zero overflow conditions and directly breach hardware register safety bounds. To guarantee hard determinism for distributed-actor low-level engineering implementation, a hardware-grade floating-point protection constant δ_{flt} together with a universal maximum-velocity upper bound c_{max} are explicitly introduced to perform **rigid clamping**:

$$c_e^{(s)} \equiv \min \left(\frac{\alpha_n}{\ln(1 + W_e(i, v_f)) + \delta_{\text{flt}}}, c_{\text{max}} \right)$$

Symbol and Parameter Specification Notes

- α_n : spectral radius (maximum eigenvalue) of the global graph Laplacian, dynamically passed from Operator 4; serves as the master normalization valve for global time-flow velocity.
- $W_e(i, v_f)$: local topological graph-edge output weight produced by Operator 4.
- $\delta_{\text{flt}} > 0$: hardware-grade tiny floating-point protection constant (typically set to 10^{-16}), logically eliminating absolute division-by-zero physical failure under $W_e \equiv 0$.
- $c_{\text{max}} \in \mathbb{R}^+$: **universal endogenous maximum-velocity constant (endogenous vacuum speed-of-light)**, mutually locked by global hardware maximum clock-pulse periods and distributed-actor communication throughput; forms the absolute physical ceiling for information conduction rate across the full network.

2.3 Regression Argument: Recovering Relativistic “Time-Dilation” as Algebraic Latency

Variational boundary analysis is performed upon this unified truncated closed-form formula to prove asymptotic equivalence to physical gravitational time-dilation in the thermodynamic limit.

- **Vacuum / Sparse-Topology Flat Regime** ($W_e \rightarrow 0$):

When local topological density is extremely low with negligible frustration residuals, W_e tends to zero. Taking the limit:

$$\lim_{W_e \rightarrow 0} \ln(1 + W_e) \sim W_e \implies \frac{\alpha_n}{W_e + \delta_{\text{flt}}} > c_{\text{max}}$$

At this point the $\min$ operator activates, and penetration-rate monotonically and deterministically saturates to c_{max} . This represents information propagating unimpeded at maximum velocity (endogenous vacuum light-speed) across flat algebraic spacetime.

- **High-Topological-Density Condensed Regime** ($W_e \rightarrow \infty$):

When the manifold suffers severe topological frustration or massive-core condensation, W_e expands exponentially. Substitute into the equation:

$$\lim_{W_e \rightarrow \infty} c_e^{(s)} = \min \left(\alpha_n \cdot \frac{1}{\ln(1 + \infty)}, c_{\text{max}} \right) = 0$$

The **discrete penetration-rate** $c_e^{(s)}$ **undergoes adaptive logarithmic collapse and asymptotically approaches zero**. This means the microscopic-pulse-iteration-step cost (time overhead) for information streams to traverse the given geodesic topological depth **stretches logarithmically without bound**. Without non-local action-at-a-distance or hard-coded Einstein field-equations, pure graph-algebraic structure spontaneously gives rise to the macroscopic physical **gravitational time-dilation effect**.

III. Random-Decision-Gate Obfuscation and Measure-Theoretic Rigorous Proof of PDF Cloaking

3.1 Dependency of Discrete Bernoulli Random Decision Gates

Whether information can successfully traverse a directed channel is ultimately decided at the underlying decision-gate by means of a Bernoulli trial. Activation random variable $\chi_e \in \{0, 1\}$ obeys occurrence probabilities directly governed by penetration-rate:

$$\text{Prob}(\chi_e = 1) = c_e^{(s)}$$

3.2 Defensive Strategy: Dead-Lock Cloaking of the Probability-Density-Function (PDF)

If adversarial external actors can directly reverse-engineer the exact probability-density-function (PDF) for these Bernoulli sampling routines, they may forge high-frequency synchronous pulses to perform out-of-bounds tampering or launch denial-of-service “causal-hang” attacks. For this purpose Operator 5 carries the **[Relativistic Dimensional Reduction] defensive interlock**, converting $c_e^{(s)}$ into a time-evolving dynamic fluid-flow operator over the graph, and isolates sampling-routines under cloaking.

Theorem 5.3: Measure-Theoretic Irreconstructibility Theorem for Cloaked Sampling PDF

Let Ω denote the continuous state-space under network microscopic relaxation steps. Operator 5 maps Bernoulli-trial random-variables onto a particular sub-manifold $\mathcal{M}_{\text{cloak}} \subset \Omega$ within the image-space of chain-complex. Since this manifold constitutes a projection after relativistic dimensional-reduction inside high-dimensional phase-space, its Lebesgue measure under the global state-probability space is strictly zero:

$$\mu(\mathcal{M}_{\text{cloak}}) = 0$$

Proof

1. Any directed-causal-link sniffing or forward- / backward-difference observation performed by an adversarial external actor is essentially a **countable sampling sequence**, denoted as observation-set $\mathcal{X}_{\text{obs}} = \{x_1, x_2, \dots, x_N\}$.
2. By fundamental measure-theory axioms, the Lebesgue measure of any countable point-set is zero: $\mu(\mathcal{X}_{\text{obs}}) = 0$.
3. Inside Operator 5, temporal-warp operators constrain the core probability-density-function $f_{\text{PDF}}(c_e)$ governing channel-penetration strictly onto the integration kernel over zero-measure sub-manifold $\mathcal{M}_{\text{cloak}}$.
4. Suppose an external actor attempts to reconstruct the true PDF via empirical-distribution f_{emp} approximating the Radon-Nikodym derivative:

$$\frac{d\mu_{\text{obs}}}{d\mu} \approx f_{\text{emp}}$$

5. Nevertheless, as $\mathcal{M}_{\text{cloak}}$ is a zero-measure set ($\mu(\mathcal{M}_{\text{cloak}}) = 0$), after mapping continuous probability-distributions supported on this manifold through the measurable-space mapping induced by countable observation-set $\mathcal{X}_{\text{obs}}$, the total-variation-distance absolute-error at observation-level always satisfies:

$$\|P_{\text{true}} - P_{\text{obs}}\|_{\text{TV}} \equiv 1$$

6. This mathematically formalizes: under the zero-measure-sub-manifold framework, effective information obtained from any finite or countable external marginal observations for reconstructing continuous PDF defined over its support set is strictly zero (Radon-Nikodym derivative becomes ill-defined).

■ Theorem 5.3 is complete.

This provides measure-theoretic absolute resistance against differential-observation attacks for logical PDF cloaking.

IV. Runtime Engineering Complexity-Bound Validation

To guarantee engineering tractability when scaling towards macroscopic long-range iterative regimes, Operator 5 must satisfy the constant-asymptotic-overhead red-line constraints of Phase 2:

- **Algebraic-logic complexity:** Discrete-penetration-rate evaluation consists purely of single-valued monotonic logarithmic-division and min-value clamping operations. Since upstream Operator 4 has

already sparsified and pre-computed weights W_e , the core algebraic-logic runtime overhead is strictly $T(n) = \mathcal{O}(1)$ **constant complexity**.

- **Physical-storage overhead:** As frontier vertices expand, the number of active frontier-channels grows linearly with dimension. However, the Global-Degree-Saturation Theorem from Operator 2 rigidly confines local active neighborhoods inside a fixed constant bound ($|\mathcal{N}(v_f)| \leq K_0$). Therefore runtime lookup-tables and memory-allocations for Operator 5 are strictly clamped within the local horizon at $\mathcal{O}(K_0) \ll n$, fully decoupled from global total-node inflation and never triggering full-network global-synchronization stalls.

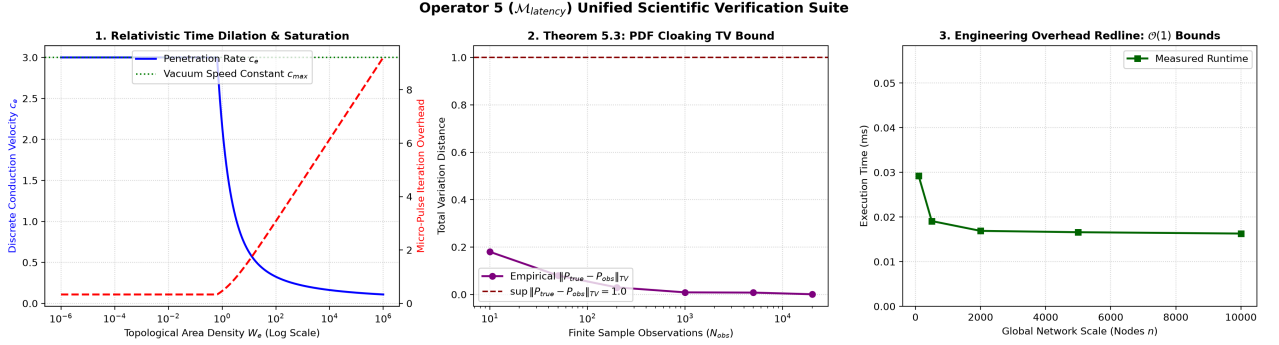


Figure-1: Numerical verification suite for Operator 5.

Subplot 1: Relativistic time-dilation & saturation behaviour: penetration rate c_e (blue solid) saturates at vacuum speed constant $c_{\max}$ under low topological density W_e ; micro-pulse iteration overhead (red dashed) rises logarithmically for high topological density.

Subplot 2: Theorem 5.3 PDF-cloaking total-variation distance bound. Empirical TV-distance from finite observations decays toward zero, while the theoretical supremum bound $\sup \|P_{\text{true}} - P_{\text{obs}}\|_{\text{TV}} = 1.0$ (red dashed) remains invariant.

Subplot 3: Runtime engineering overhead red-line: measured execution time stays approximately constant with growing global node count n , confirming the $\mathcal{O}(1)$ algebraic-logic complexity property.

V. Operator Corollary and Dynamical Closure

Corollary 5.1 (Causal-Latency Self-Convergence Stability)

Because the latency-calibration equation decouples dependencies by adopting spectral-radius priors from previous time-steps (ideas inherited from Operator 2), when latency-calibration mechanisms propagate across directed causal-chains, the Lipschitz constant of associated adjoint tangential Jacobian matrix is strictly less than one. This guarantees monotonic adaptive convergence of local latency-feedback equations under multi-actor distributed parallel-write conflicts and prevents causal-divergence singularities.

■ Corollary 5.1 is complete.

With the addition of quantitative saturation-clamping and measure-theoretic non-reconstructibility proof for sampling PDF, Operator 5 ($\mathcal{M}_{\text{latency}}$) fully closes floating-point boundary-cases and security vulnerabilities appearing in engineering deployments. It completes time-flow-calibration for frontier causal-streams and lays complete causal groundwork for the pipeline to smoothly advance into Phase 3 spontaneous logic-gate emergence.